SIX SIGMA IN EDUCATION

Certification Planner LLC

Proprietary and Confidential

Contents

DEFINE Phase 2

MEASURE Phase..... 4

ANALYZE Phase..... 8

IMPROVE Phase..... 18

CONTROL Phase..... 23

DEFINE Phase

Problem Statement: Improve scores of 3rd year Engineering students

Background: Currently we see challenges in students both in terms of depth of knowledge and marks secured. This impacts their ability to score adequate marks to meet the minimum cut-off and even when they get marks, it becomes hard for them to clear the interviews.

Project Scope: For this project, we have decided to keep the project scope to Third year Engineering Students of computers division.

Stakeholders:

1. Engg. Students
2. Faculty
3. Training & Placement Officer (TPO)
4. Parents
5. Recruiters of companies
6. Hiring Managers

Voice of the Customer (VOC):

The starting point of Six Sigma project is to identify who the customers are and what do they need. This is captured in the Voice of the Customer (VOC).

We interviewed multiple stakeholders from each category to understand what do each of them want and here is what we found.

We ensured that we interviewed 5 people from each category of stakeholders and then consolidated what they have mentioned without filtering / weightage.

#	Stakeholder	VOC
1	Engg. Students	1. Need more depth in the subject 2. Understand the weak areas in each subjects and focus on those 3. More practical to understand the subject 4. How to apply concepts in real world problem solving 5. Employability focused education 6. Help in understanding expectations from corporates
2	Faculty	1. Need to find out weak areas of students to help them understand well 2. Ensure students attend classes 3. Understand weak areas and improvise teaching 4. Improve net knowledge level of students

3	TPO	1. More students meet the cut-off criteria 2. Good interview performance 3. Make students more capable of getting job
4	Parents	1. College provides good education 2. Improve knowledge of their wards 3. No additional/external coaching 4. Employable
5	HR/Recruiters	1. More students getting selected from recruiting rounds 2. Knowledge of latest technology 3. Ready for employment 4. Get new fresher employees productive in less time
6	Hiring Managers	1. Adequate depth 2. Ready to start working 3. Less lead time / training needed to perform
7	Management	1. Improving overall teaching and learning standard at the institute 2. Attract good number of students 3. Achieving close to 100% admission 4. Attract good number of employers 5. Achieving close to 100% placements
8	Community	1. Overall standards of education improves 2. International level education in India 3. Less need to go out of India to learn 4. Less unemployment in the society

VOC Analysis

Based on the VOC collected, analyze them and understand what do the customer / stakeholder really need. Some of these requirements are specific asks and others are generic.

As you can see there are several needs as stated by stakeholders. These will result into several Green Belt and Black Belt projects.

For this project, we will focus on the 2 specific needs and start collecting the information and move to the next phase and run the project through all the phases.

The areas we will focus would be these:

1. Understand the weak areas in each subjects for every students and provide the insights to the students and to the faculty (Students' Needs). This will help students understand their weak areas and improve. Similarly, the faculty can help the students to improve.

2. Need to find out the areas where majority of the students have not done well and these could be potential areas of improvements for the faculty to improve herself and/or deliver additional lecture on the same (Faculty's Needs).
3. These needs to be done for each subject for a particular class (for this project we will take 2nd year students).

The sample to be selected was in question, should we look into final year or third year or second year. Final year obviously would be very late, third year could be a good target to begin with. The concern we have with 3rd year is that it could be too late to get all the benefits. The learnings from these can only be applied for about a year.

After couple of rounds of discussion, we concluded that it would be good to start at 2nd year and also do it in 3rd year to get continued benefits. These can also propagate back to 1st year.

For this project to be successful, we would need to evaluate test/exams scores as starting point from 3rd year – as these students need more "immediate" help than 2nd year students.

Project Scope:

Using Six Sigma analyze weak areas of students & teacher and help them understand what needs to be done to bridge the gap.

This project will also give details on how the Lean Six Sigma methodology can be used to understand and gain insights into education domain to explore opportunities and solve problems effectively.

MEASURE Phase

To start the measurement of the current performance, select the most recent exam / test the students have had.

As part of the data collection, we first need to do measurement system analysis (MSA).

We need to understand the best way to capture these information.

After discussing and brainstorming with students and teachers, we found the following:

1. Students get to know the marks in each subject (without details on their performance on each questions)
2. There are some teachers that give breakup of marks for each question for the students
3. There are few teachers that highlight the mistakes / put comments on what is the problem and why/where the student lost marks

After further discussion and analysis, we found that none of the cases, teachers capture marks of each students by the question number. Also, they do not keep records. It becomes difficult to find patterns in performance and weak areas of the students / faculty.

Data Collection:

We defined the measurement system, where for every test conducted, teacher will capture details for each students for each subject, each question.

We asked teachers to start capturing data for each of the students for each subject, each questions the student appear.

While looking at the data, we also observed that the students have option to select 4 questions out of 6 given. That adds to the complexity of finding out if the student skipped the question or was not aware of the answer.

Typically in situations like this, we can assume that the data would be randomly distributed. i.e., each question would have equal likelihood of getting selected by the student to answer.

When we spoke to the students, they mentioned that they typically start with the first question and continue with other questions and only when they are not sure – they will skip that question and usually will end up skipping the last question.

Also, there is another factor if somebody have skipped a question or have received less marks in a particular question or have not appeared, it could also be because they may have ran out of time.

For the purpose of this analysis, we would use the following operational definition.

Operational Definition: Marks scored for each question by each student will be considered as their knowledge level. Any question skipped or not appeared by the students due to lack of time or due to negligence are not treated differently – they are considered as not enough knowledge.

The 2nd level of analysis can be used to understand this from each student after they receive their scores to understand their remaining parts.

We need to capture metadata from the teachers on the background of each of the question. The data that we need is what syllabus/topic the question covers and the type of question. This would be needed to understand depth of syllabus coverage by the student and also help in providing recommendation on what they need to focus on.

Here is the data collection sheet that was created.

The column name captures the question number along with the maximum marks possible by a student if they appear those question.

Here is the snapshot of the data collection sheet.

Engineering College											
Department of Information Technology											
Marksheet for MID Term 2013-14 (Even)											
Class/Div: TE (A)			Subject: Computer Networks					Name of Faculty: Riah			
Roll No.	Set	Name of Student	Q1(10)	Q2a(5)	Q2b(5)	Q3a(5)	Q3b(5)	Q4(10)	Q5(10)	Total	Comments

Column “Roll No” Captures the roll number – unique identifier of the student

Name of the student – we have kept the first name of the student

Other columns are capturing the question numbers. Some of the questions are for 10 marks and some are for 5 marks. These are mentioned in the brackets

The teacher will fill out this sheet for each of the student.

Next steps is to get the data from each of the student’s answer sheet evaluated by the teacher.

The raw data is available in the enclosed excel workbook.

The data is collected and presented.

First, do the basics – get what the teacher is expected to do. Calculate sum of marks achieved by each students.

Here is a snapshot of the marks for students for each question.

Engineering College												
Department of Information Technology												
M arksheet for M ID Term 2013-14 (Even)												
Class/Div: TE (A)				Subject: Computer Networks					Name of Faculty: Riah			
Roll No.	Name of S tudent	Q1(10)	Q2a(5)	Q2b(5)	Q3a(5)	Q3b(5)	Q4a(5)	Q4b(5)	Q5(10)	Q6(10)	Total	Comments
1214001	ALEX	9	5	4		3		4	5		30	
1214002	JOSEPH	8	4	4		3	4	4			27	
1214004	DAVID	10	2	4		5	3	4			28	
1214005	MICHAEL	9	5	5		5		4		3	31	
1214006	JOHN	10	5	5	1	5		3			29	
1214007	WILLIS	8	5	3	5	4	1	4			30	
1214008	WILLIAM	9	5	3		4	2	4			27	
1214009	RICHARD	9	3	4	1	4	4	5			30	
1214010	CHARLES	8	4	5		5	2				24	
1214011	CHRISTOPHER	9	5	5		5	3				27	
1214012	DANIEL	9	5	5		4					23	
1214013	PAULA	10	5	5	3	5	4	4			36	
1214014	MARY	9	5	4		5	4	5			32	
1214015	PATRICIA	9	5	4		5	5	5			33	
1214016	JENNIFER	8	5	5		4					22	
1214017	LUCY	7	5	5		5		4			26	
1214018	LISA	9	5	4		5	4	4			31	
1214019	NANCY	7		3	2	2		4			18	
1214020	SANDRA	8		5	5	5		4			27	
1214021	CAROL	8	5	4	5	5	1	3			31	
1214022	DONA	9	5	4	5	3	2	4			32	
1214023	MICHELLE	9	4	3	5	4	4	4			33	
1214026	SARAH	9	5	4		5					23	
1214027	KEVIN	9	5	5		5		4			28	
1214028	KIMBERLY	9	5	4		4		4			26	
1214029	DEBBY	10	5	3	2	4	4	4			32	
1214030	GEORGE	9	5	3	3	4		4			28	
1214031	ROSS	9	5	5	3		2	4			28	
1214032	KENNETH	8	4	5	3	5		4			29	
1214033	SUSAN	10	5	5	5	5		4		3	37	
1214034	MARIA	9	5	5	5	5	5	5			39	
1214035	RONALD	10	5	5	5	5		4			34	
1214036	DOROTHY	9	2		1	5		4			21	
1214037	ELIZABETH	9	5	5		5		4			28	
1214038	LINDA	10	4	5	5	5	5	5			39	
1214039	MARK	9		4		4		2			19	
1214040	THOMAS	6		5		4		4			19	
1214041	EDWARD	6		5		5		3			19	

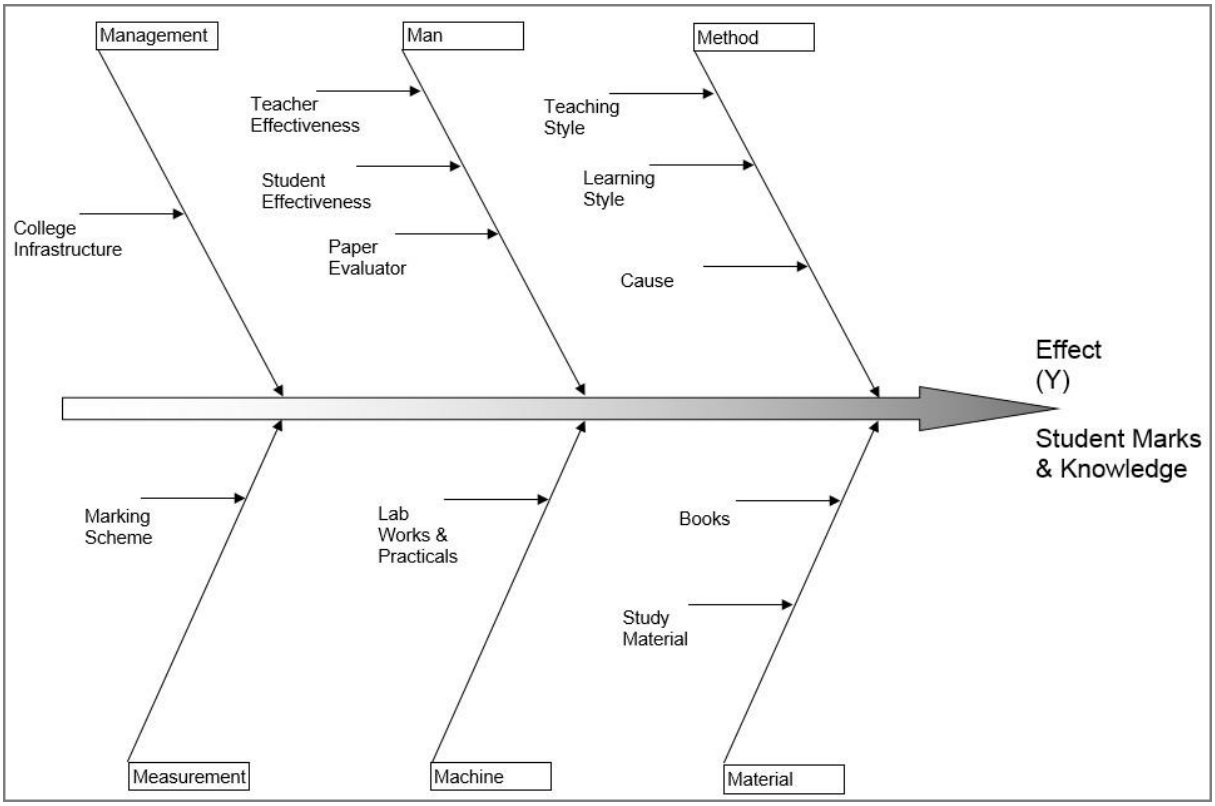
You would work with the teacher to understand the mapping of these questions to the content - syllabus / chapters.

ANALYSE Phase

In the Analyze phase, we start looking at the data and analyses using multiple ways to gain insights.

The analyses approach needs to be brainstormed and appropriate tools / approach must be used to understand how students have scored and understand their gaps and opportunities for teachers to improve.

We will first create a cause and effect diagram to understand well what activities will have an impact on the result.



Step 1: Understand how students have done in each question

1. Take one question at a time and understand the overall spread of marks by the students.
2. Find out how many students did not appear the question (either they chose to use that for option and/or did not knew the answer) – for those questions, count the marks as zero.
3. Calculate the average marks achieved by the students for each questions.
4. Wherever the students have scored less than average marks for the question, those students should consider an opportunity for improvement.
5. The hypothesis here is that even if students have got good marks, they would be compared against the class and then find out opportunities for improvements.
6. You can have some exceptions for these, like…

   (Please note: The number 80% was decided based on discussion with teachers and students – who mentioned that 80% is good cutoff to demonstrate proficiency on the topic/question. Depending on the domain and complexity of the project, you can choose)
7. You may choose to keep a bar of 80% as a cut-off (i.e. even if the students have received less than average, if he/she have scored above 80%, you can decide if you want to keep that particular question as improvement area for the student).
8. At the same time if the average score is very low – then, it might make sense to have more student look at the improvements.
9. Collectively you can choose to have any student who have scored below 80% to look for improvement.
10. There is one additional aspect to be considered – that is: the low score could be because of teacher not teaching the topic in good details.
11. Do this for each of

Step 2: Understanding the topics/questions where majority of the students have not done well

1. There are couple of signals you can get from the marks secured by each students for the questions
2. Find out how many student have not appeared the question. Any question that is not answered by more than 10% of the students is considered an opportunity for improvement for the teacher for that topic (Please note: This 10% can be set to 20% as appropriate, based on the condition and needs)
3. The next thing to consider is the average marks achieved by the students. If the average marks achieved is less than 60%, then – this is definitely a topic which majority of the students are not finding it difficult. (Please note: the 60% is an KPI to assess average

marks across students for a particular topic/question to help understand the overall performance)

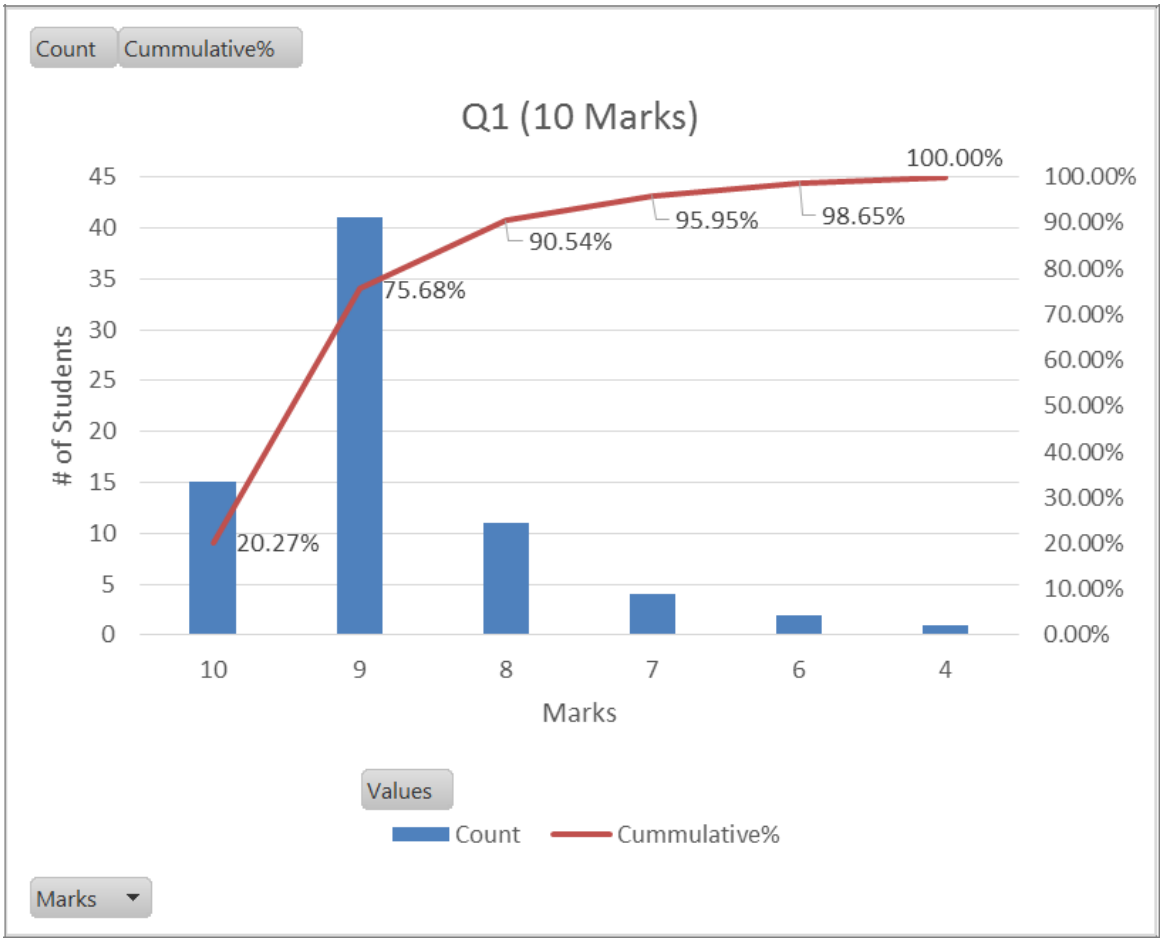
4. If more than 25% students receive less than 50% marks – then it is considered as an area where students have not received adequate knowledge / depth

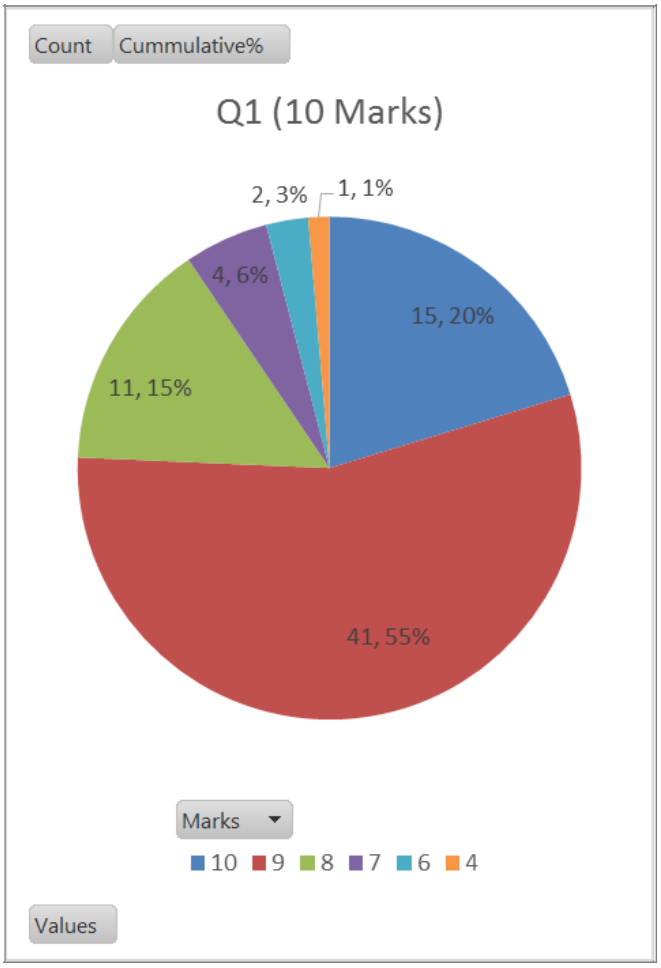
Data Analysis for Question 1

We looked at the data and now doing data analysis

Here are the snapshots of some of the data analysis

Figure 1: Pareto Chart showing distribution of students marks for question # 1. These are sorted in descending order with cumulative %.





Observations: You can observe that 20% of the students have achieved full marks of 10. More than 90% of students have achieved 80% marks or more marks.

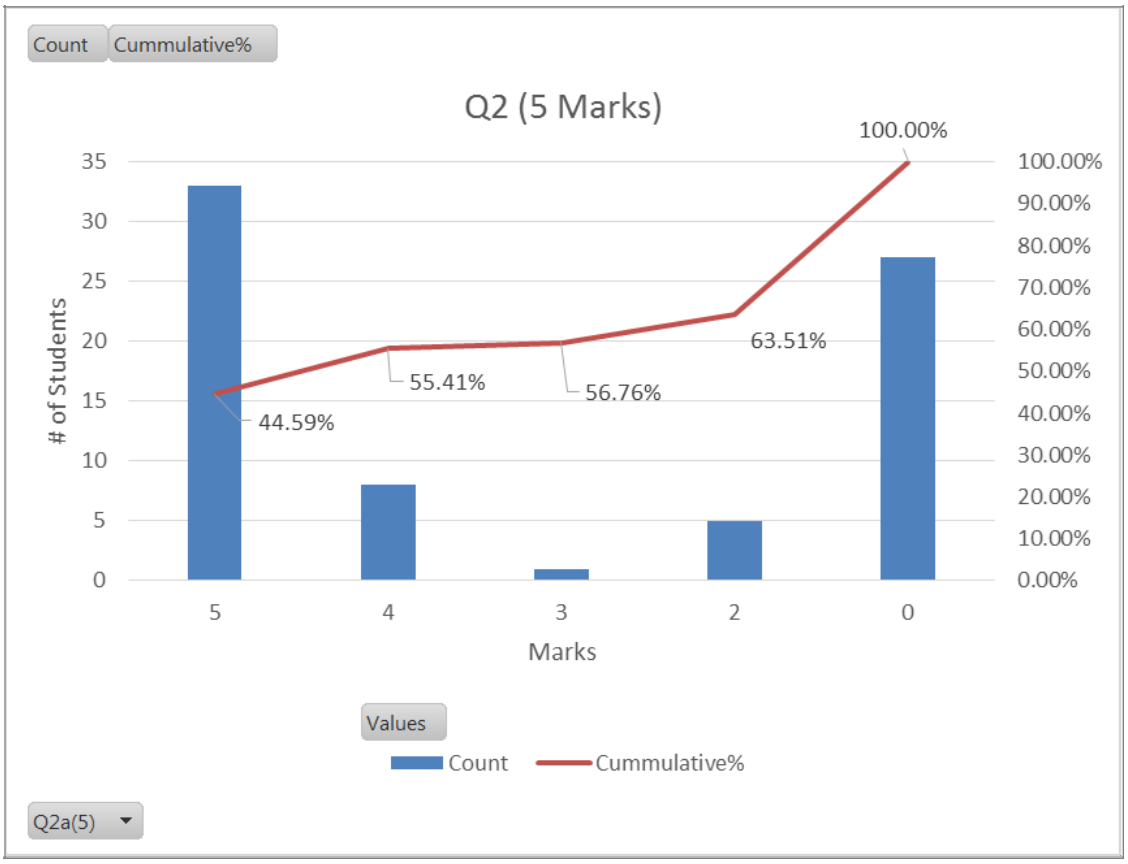
- Mode would be 9 marks
- Mean is 8.8 marks
- Median is 9 marks
- 55% of the students have secured 9 marks

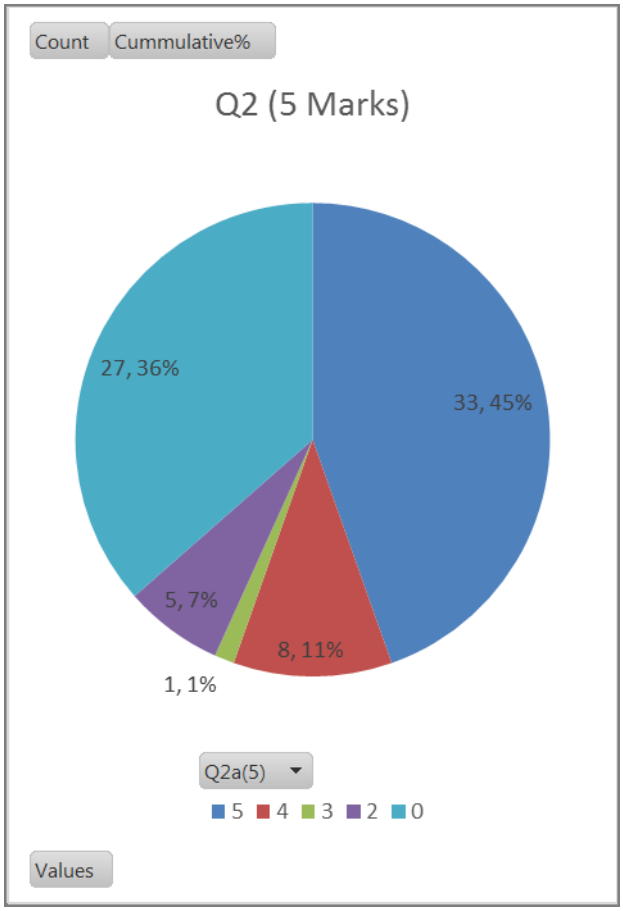
There is only 1 student (out of 74 students) (1.4%) that has secured less than 50% marks.

Conclusion:

It appears that the teacher was able to deliver the content topic very well – with 90% of the students getting more than 80% marks.

Data Analysis for Question 2





Observations: You can observe that 56% of the students have achieved 3 or more marks (60% of more). i.e., 44% of the students have received 40% or less marks. This would be an area of improvement for both students and teacher. Chances are that the teacher would probably not have delivered the topic well or the understanding of the topic is not good with the students

- Mode would be 5 marks
- Mean is 2.8 marks
- Median is 4 marks

The huge delta between Mean and Median tells you something important. It says some students have got this topic very well 44% students get 100% marks (5 marks). And a staggering 36% folks have got zero or have not appeared this question.

One thing to explore here would be to see if the students opted to leave out this question.

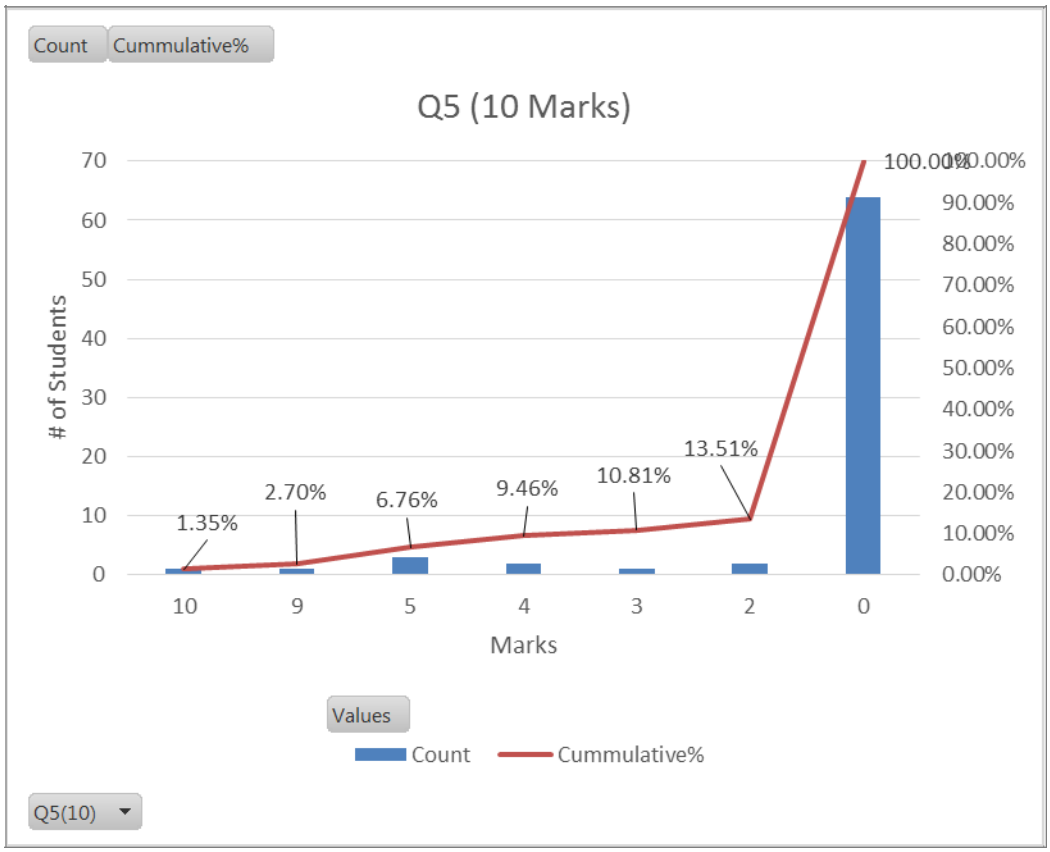
Next level of analysis shows that people who have scored zero marks in question 2a have scored very well in 2b. Which proves that the students have not opted to leave this question out.

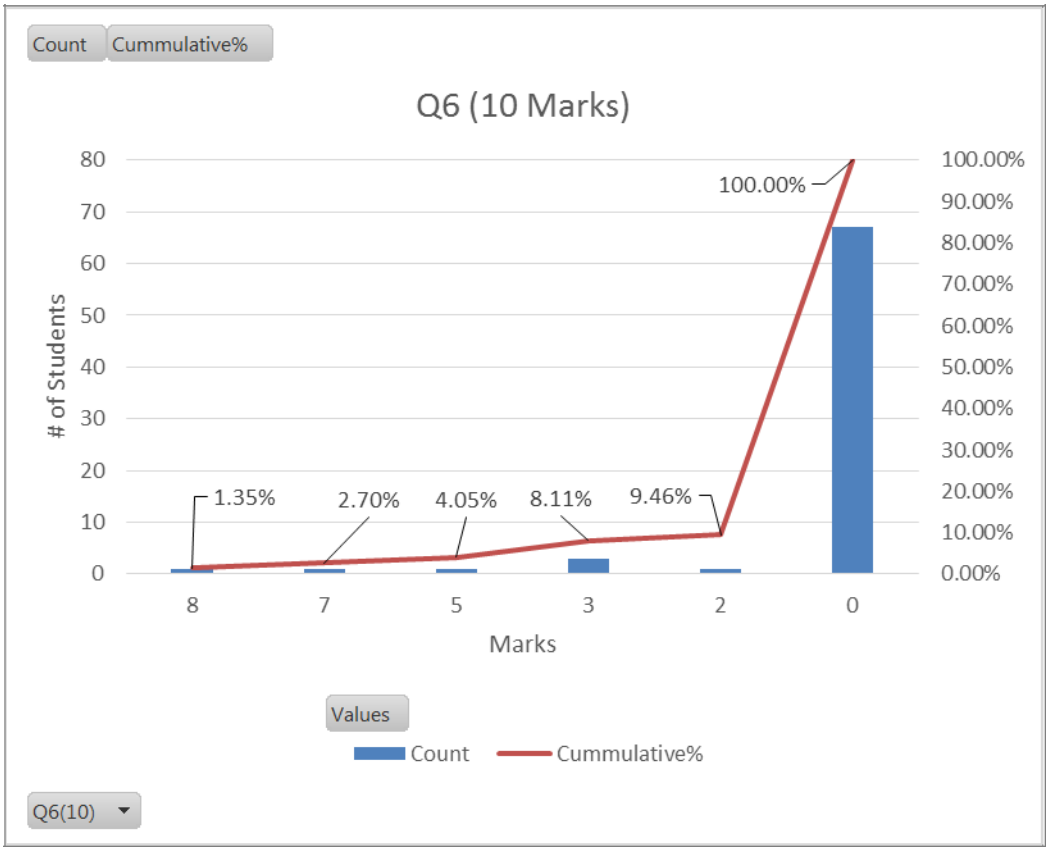
Conclusion:

It appears that the topic is difficult for a good % of students. The teacher probably can focus on providing the guidance for the students on the topic related to question 2a.

Now, let us jump to the last question.

Data Analysis for Question 5 & 6





Observations: This question is the one where majority of the students have opted to leave the question or have secured zero marks.

- More than 86% of the students have got zero marks in question # 5.
- More than 90% of the students have got zero marks in question # 6.

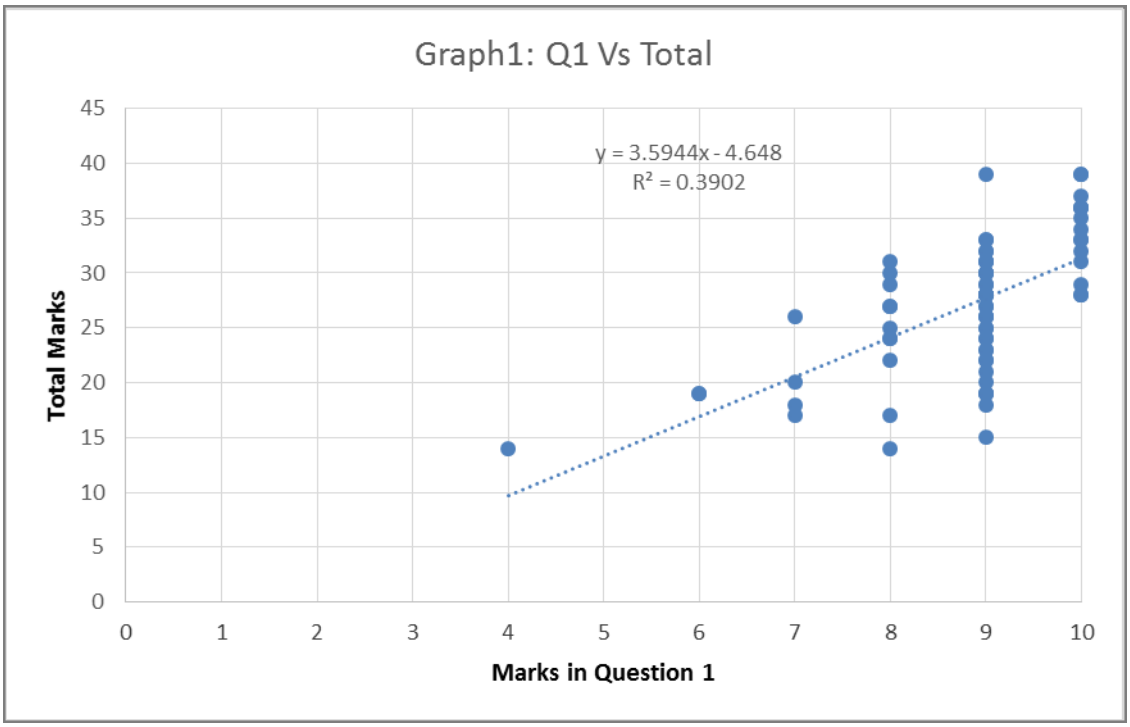
Both of these questions are clear topic for the teacher probably can teach again.

These type of analysis to be done for all the other questions to gain insight.

When doing the analysis for individual students

Correlation analysis was performed to see if question # 1 was any indication for the overall performance of the students. For this scatter plot was used.

Here is what was observed:



There is No Strong Correlation that stands out. But there is an overall pattern.

- People that have got 10 marks in Q1 are the one that have got more than 28 marks in total (average 34).
- People that have got 7 or less marks in Q1, have not scored more than 20 (except for 1 person)

Here is the summary of analysis

Q1	Majority of the students were comfortable with Q1. Only 1 person got 4 marks out of 10 rest all the 73/74 students got 6 or more marks.
Q2a	Only 2/3rd students appeared this question. Overall people did well in this question with an average score of 4.5/5 marks. Only 11% of the people got 2/5 or less
Q2b	Majority of the students were comfortable with this question. Only 3 person got 2/5 marks or less. On Average students got 4.4/5 marks for this question
Q3a	This is the weakest question - only 21/74 (28% students) appeared this question. And for people that appeared this question have got only 3.4/5 marks average and 33% of the students getting 2/5 or less marks in this question
Q3b	Majority of the students were comfortable with this question. Only 2 person got 2/5 marks or less. On Average students got 4.4/5 marks for this question
Q4a	While 51% of the students appeared this question, the average marks was not very high, it was 3.6/5 marks with 21% students getting 2/5 or less
Q4b	This question was attempted by 3/4th of the students and on average it was on the lower side with 3.9/5 marks. 7% of the students who appeared this question got 2/5 or less marks. Overall low scoring question
Q5	Only 10 students appeared for this question, out of these 2 students got 9 or 10 marks - these are the students who are only in top 3 scorer. Remaining students got less marks in the range of 2-5 marks out of 10.
Q6	This is the weakest question - only 7/74 (9% students) appeared this question. And for people that appeared this question have got only 4.4/10 marks average

Top 25% Students	
	Only 1 students from the top 25% students did not appear for Q2 - rest everybody appeared the questions.
	Around half of the students from top 25% did not appear question Q3a but have appeared for Q3b (except for 2 students who decided to skip both Q3a and A3b).
Bottom 25% Students	
	Majority of the bottom 25% of the people have appeared Q1, Q2b and Q3b. Several attempted Q4b as well. That Probably shows these questions were easy or well taught in the class.

Legends:

Overall knowledge depth acceptable
Some improvements can be done
Needs improvement
Serious issue expected – to be fixed immediately

IMPROVE Phase

The next step is to start improving the process.

Start with conducting brainstorming with the individual stakeholders and understand how things could be improved.

Please note, since this is a sensitive topic, it would be good to start with the teachers and take their views on how to go about improvement after validating the report.

Typically, for this type of project DoE (Design of Experiments) is good.

Here, we ask the teacher to try out different combinations for the questions and see what impact it has. To understand all the sources one need to understand the following aspects and see what improvements are needed in each of these.

1. Topic
2. Content
3. Study materials
4. Teaching style
5. Time available
6. Practical
7. Type of question
8. Depth of answer

Select a topic and work with teacher on improvising the topic and conduct a refresher training and give the same question that was in the exam/test and evaluate the performance. If we observe that the performance has improved – it means our pilot is working and similar approach can be applied on the other questions and topic.

Work with teachers and evaluate corrective and preventive solutions for each of the areas and confirm what is working and what is not working and propagate the learning to other teachers.

It was found that majority of the teachers liked this approached and there was a need to make it more systematic and early feedback rather than waiting till the end to get the information and then acting in a reactive fashion.

We had 4 brainstorming rounds and finally came out with the following suggestions that all of them were willing to implement.

Improvement suggestions:

1. Focus on adequate depth on the topic.
2. Every day, reserve last few minutes to quiz students on their knowledge level.
3. Have early and regular tests / feedback loop to understand if students are facing any challenge in any of the topic.
4. One of the important improvements that came up was to announce ahead of time what chapters / topics would be covered on what dates/weeks and encourage students to read/lean about the topic before hand – that way there is right set of expectations and can ask questions.
5. Majority of the people involved in the projects, stakeholders and also the students found this useful and potentially effective.
6. This will make systematic improvement rather than reactive.
7. We need to validate if this is going to work. So we need to do an experiment.

Experiment

To validate the improvements, we asked teacher to teach a topic in one class the traditional way and the same topic was given to the other class ahead of time and asked students to read and prepare ahead of time.

Class Set 1:

1. Inform the students one week in advance that the particular topic will be taught.
2. Ask students to read the material and also refer to any reference material like YouTube / websites / etc.
3. Deliver the lecture on the pre-determined day.

Class Set 2:

1. Select the same topic as done for Set1.
2. Select n topic, do not inform the students on what topics will be taught / when.
3. Teach the students in the traditional way.
4. Teach the student on the agreed upon day (within +/- 1 day of when it is delivered to class set 1).

On 10 day after the class, give the same test/assessment to all the students (Class Set 1 and Class Set 2) – together. The topic of the test/assessment would be the same topic that was taught with no optional question. The focus of the test would be gain understanding of the knowledge / depth of students for the topic.

Mix up all the papers and evaluate the papers without knowing if the student belong to set 1 or set 2. After all the papers are evaluated, put the marks against the roll numbers. Identify the data set and segregate set 1 and set 2.

Then do comparison of the data and see if there is significant difference in the data. You can look for the following:

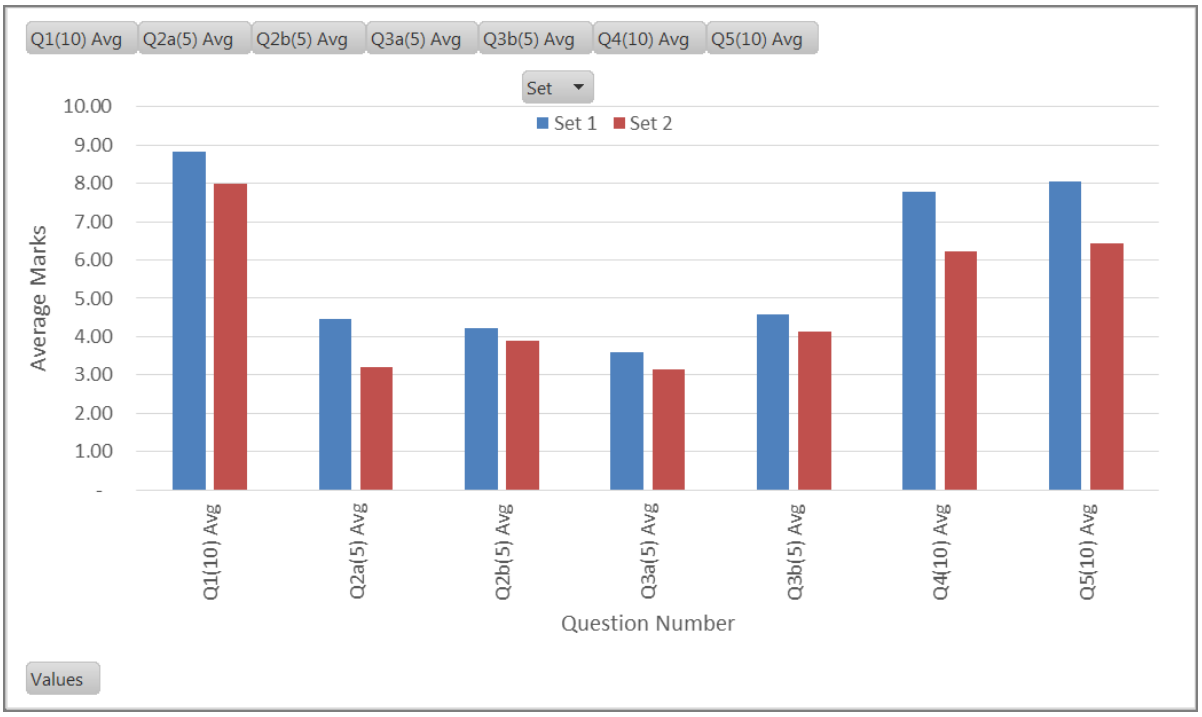
1. Mean Marks secured
2. Spread of the data
3. Std. Deviation
4. % of students in each quartile
5. Compare each of the data set with the other set

You can use paired t-Test for doing before and after analysis.

The purpose of this is to see if the new improvement is working and showing significant enough improvements.

The data set if available in the excel worksheet

Question #	Set 1	Set 2	Overall Average
Q1(10) Avg	8.84	8.00	8.42
Q2a(5) Avg	4.46	3.19	3.82
Q2b(5) Avg	4.22	3.89	4.05
Q3a(5) Avg	3.59	3.14	3.36
Q3b(5) Avg	4.57	4.14	4.35
Q4(10) Avg	7.78	6.22	7.00
Q5(10) Avg	8.05	6.43	7.24



From the data is appears that Set 1 definitely performed better than Set 2 across all the questions.

Some of the questions have minor difference and some have very high difference. But it is clear that Set 1 have performed better than Set 2 in each and every question.

Question 3a still appears to be weak for both Set 1 and Set 2. This can be further analyzed by the teacher to understand the gaps and opportunities for improvements.

Remaining questions have shown good overall progress with Set 1 achieving 80% or more marks on average by the students.

Now let us use Paired t-Test and see if the difference is statistically significant.

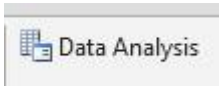
Null Hypothesis: there is no difference in performance of students in Set 1 and Set 2. i.e., the improvement that is done will not result into any impact.

You can do the test for each of the question for understanding this in granular details. But, since we are looking overall performance of students on the particular topic you can choose to start with the total marks secured by the students.

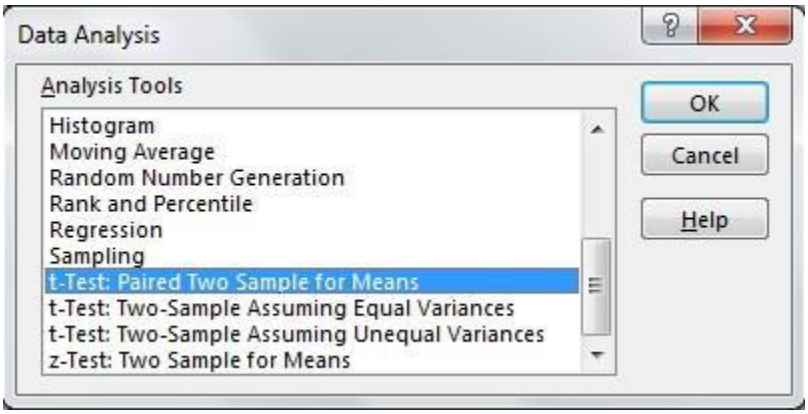
For this, we have chosen to do the test on the total marks by each student in Set 1 and Set 2.

For using the Paired t-Test, follow the following steps:

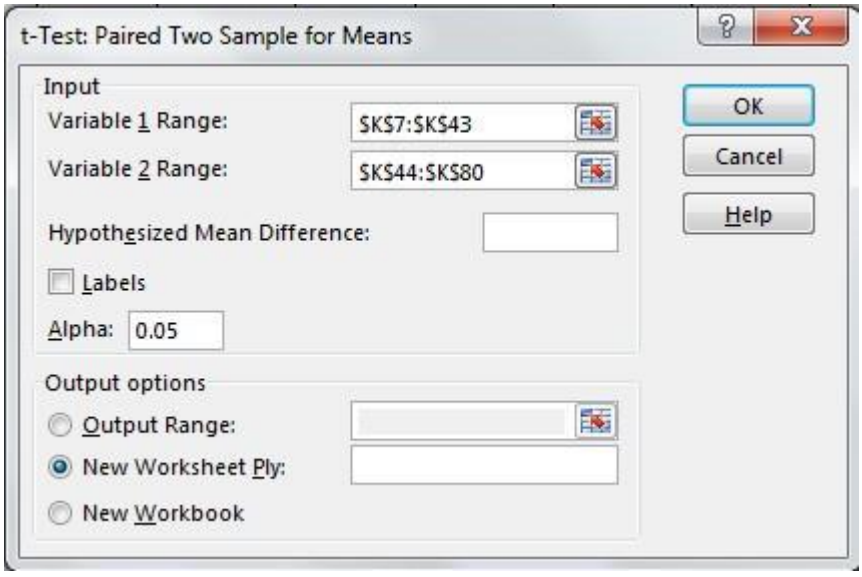
1. Go to Data Tab and click on “Data Analysis” button.



2. From the list, select Paired T-Test to test two sample for means.



3. You will see the following box:



- For Variable 1 Range – select the total marks secured by students from Set 1.
- For Variable 2 Range – select the total marks secured by students from Set 2.
- Keep the alpha value as 0.05.
- And click OK.

4. You will get the t-Test details as shown below:

t-Test: Paired Two Sample for Means		
	Variable 1	*Variable 2*
Mean	41.51351351	35
Variance	20.47897898	22.33333333
Observations	37	37
Pearson Correlation	0.215612861	
Hypothesized Mean	0	
df	36	
t Stat	6.836134459	
P(T<=t) one-tail	0.000000027	
t Critical one-tail	1.688297714	
P(T<=t) two-tail	0.000000054	
t Critical two-tail	2.028094001	

Observations & Analysis

From the Paired t-Test, you can observe that the overall average of people in Set 1 is 41.5 compared to Set 2 average of 35.

The Variance of Set 1 is 20.48 and is lower than Variance of Set 2 which is 22.33

We observe that the p-value for one-tail and two-tail is less than 0.05

As per null-hypothesis, if the value of p is low, null must go.

Hence, we reject the null hypothesis.

Based on the data, we now confirm that there appear to be noticeable change in the performance of the class of Set 1.

Based on this observation, we can continue this model and roll it out to other classes.

CONTROL Phase

In the last phase CONTROL phase – we need to ensure that the updated process is followed and the feedback loop is inbuilt.

The purpose of the control phase to have process and outcome in check i.e., ensure that the improved process is followed and the results are consistently better than what was before implementing the six sigma project.

A working team need to be set that will monitor the improvements implementation and output performance on a regular basis.

The goal is to make it part of the process such that in future even monitoring is not needed.

While it is important to have a team that can do the monitoring if the process is followed, real success can only be achieved when the teachers make it part of the process.

Here are some of the control (sustain) measures on how this would be made part of the system by individual teachers and sustain the gains:

1. Teachers speak with students on regular basis and see areas of improvements and do course correction.
2. For every topic, teachers will give 3-7 days heads-up for students to learn on their own.
3. Students will use the data for self-improvements.
4. Anytime a student get less than 80% marks in any subject – should focus on self-improvement.
5. Teachers would focus on the topic if the average marks are below 60% (this is decided based on the feedback from the teachers). This number can be further enhanced as the teachers gain more confidence in this format of teaching.